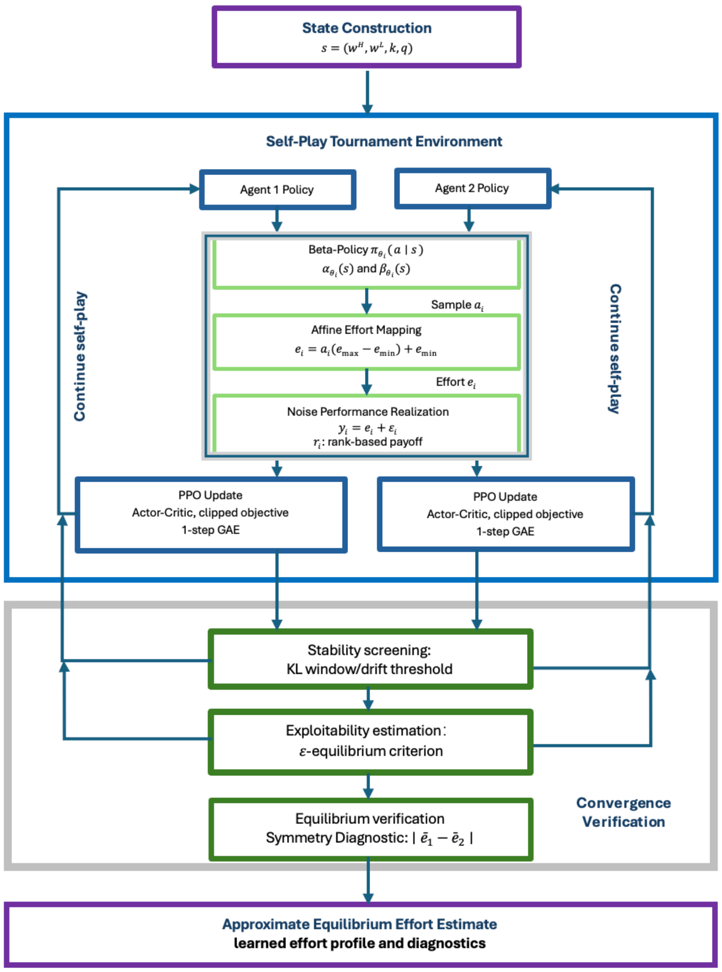
1- Framework: state dimension = 3/4

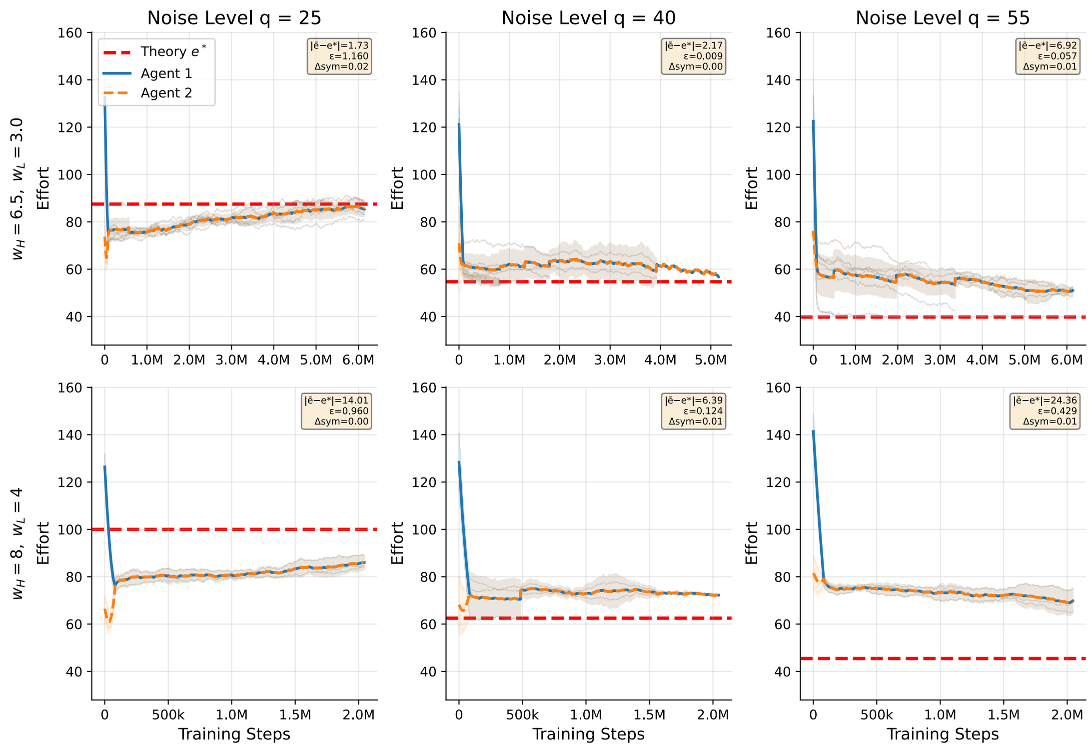


2- q=35 替换 q=25，保留 q=25 结果做分析

3- number of seeds

4- Entropy？

5- Convergence of TEL-PPO under Increasing Noise



a. 标签$(w_H,w_L) = (6.5,3)$ $(8,4)$，加上 k 值，另外 更清晰些，放在上方或加重

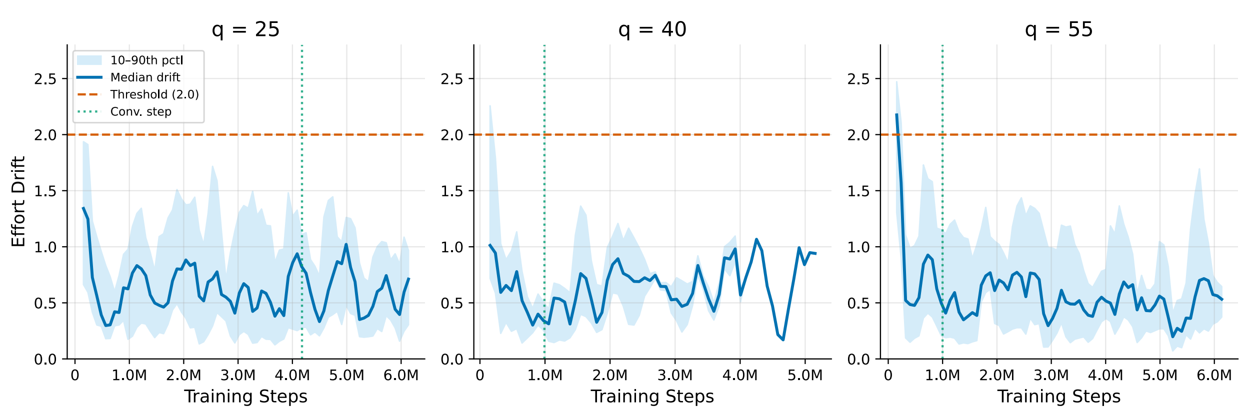
b. 阴影再淡一些

c. 加上一个 vertical line：first update satisfying the convergence criteria

d. 考虑去掉 error box

e. 这里的ε 具体指什么？

6- Effort drift across noise levels.

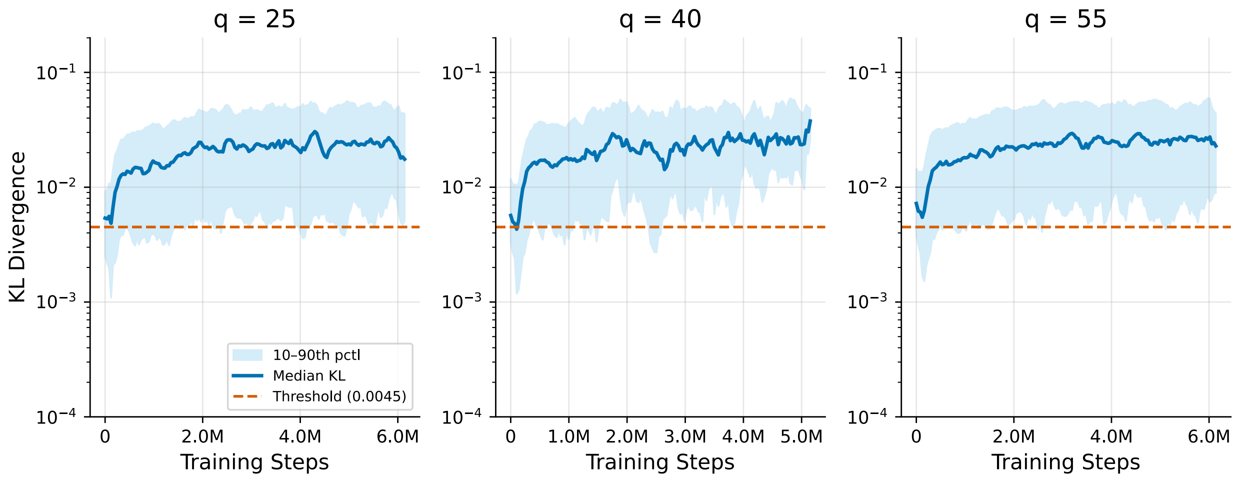


a. 阴影再淡一些

b. Threshold 粗一些

c. Legend: Median drift; 10-90% interval; Drift threshold（2.0）; Detected convergence step

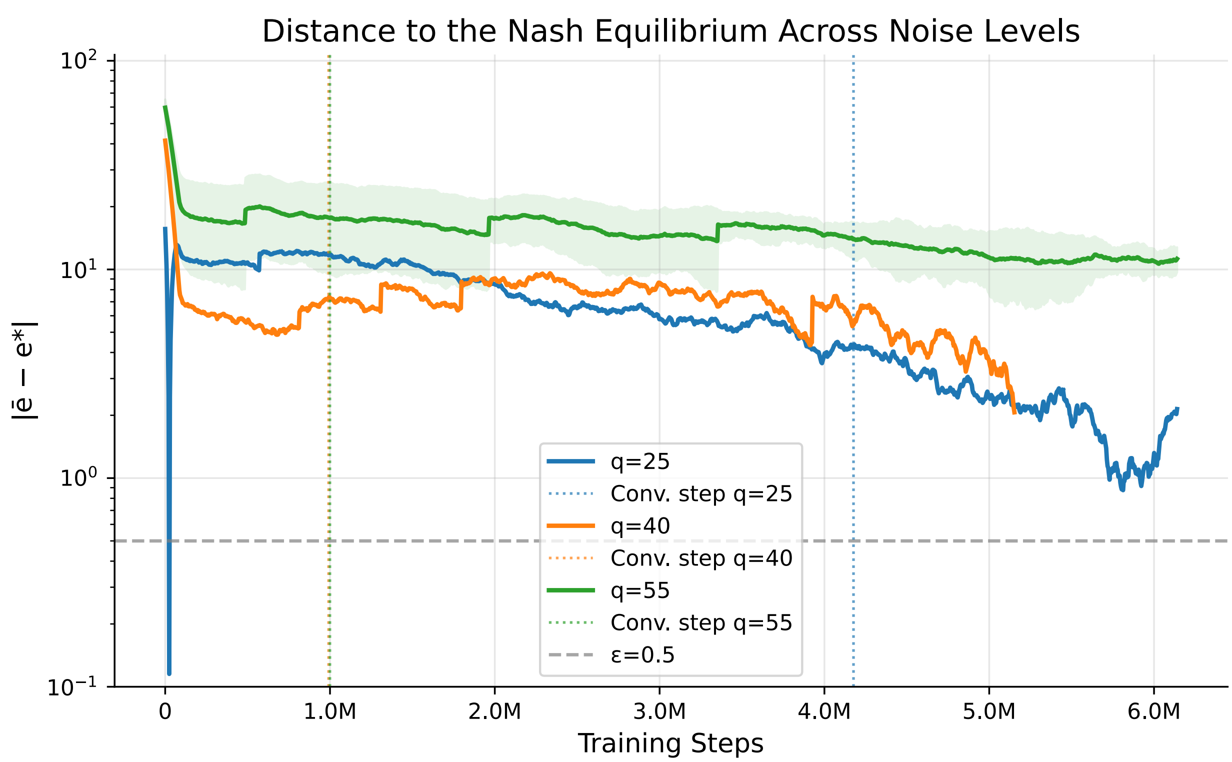
7- KL Divergence across Noise Levels



a. Legend: Median KL; 10-90% interval; Reference threshold

b. 阴影减淡

8- Convergence Error to the Analytical Equilibrium



a. Name: Convergence Error to the Analytical Equilibrium

b. y: Equilibrium error $|\bar{e} - e^*|$

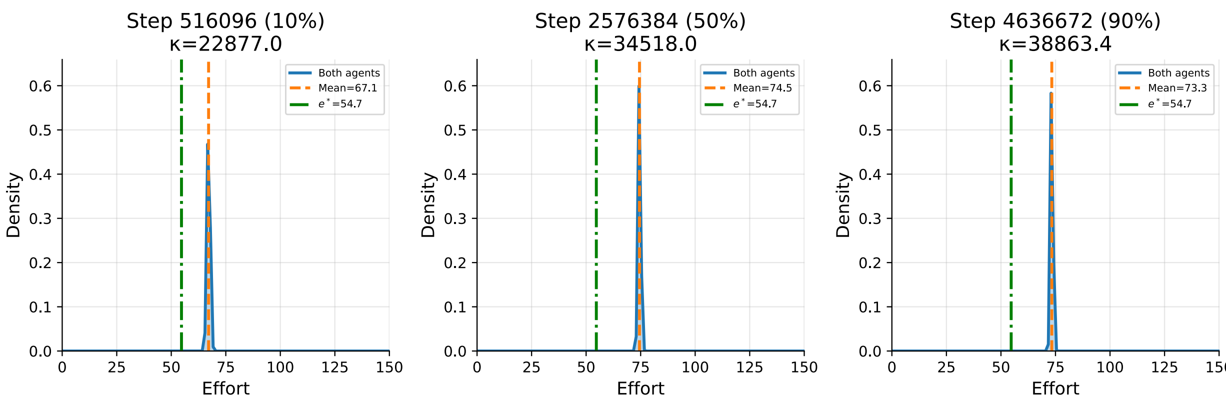
c. Target error threshold (ε = 0.5)

d. Vertical line: Detected convergence step （细一些）

e. 阴影减淡

f. Legend: q = 35;q=40;q= 55; Target error threshold; Detected convergence step

9- Evolution of the learned policy distribution over training

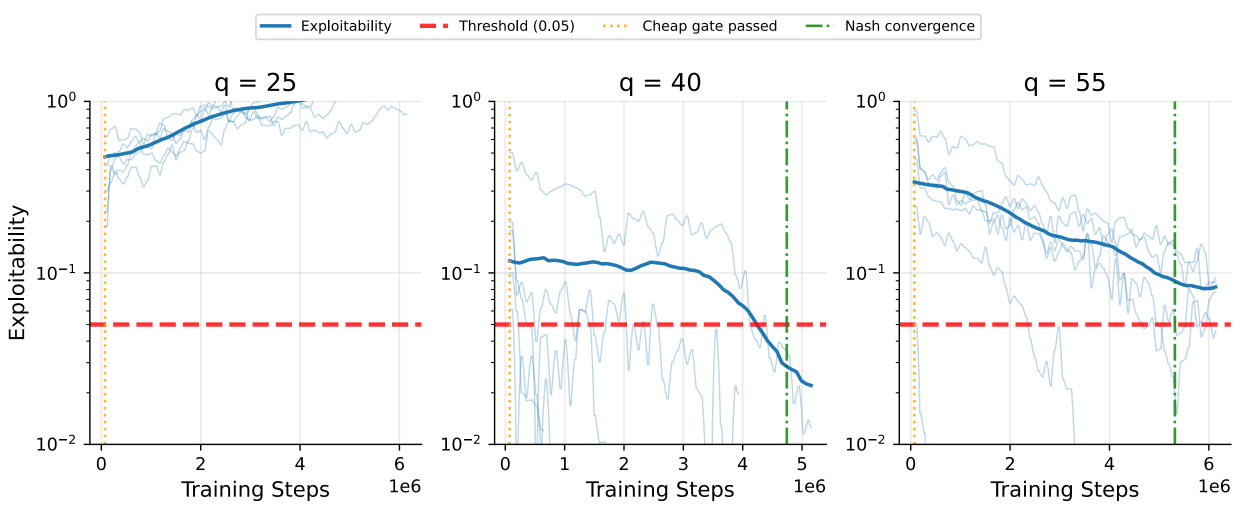


a. q = 40,

b. (w_H,w_L)=(6.5,3)

c. 有没有 error 小一点的图？（没有的话，我在考虑删去）

10- Figure 6a Exploitability and approximate equilibrium verification across noise levels



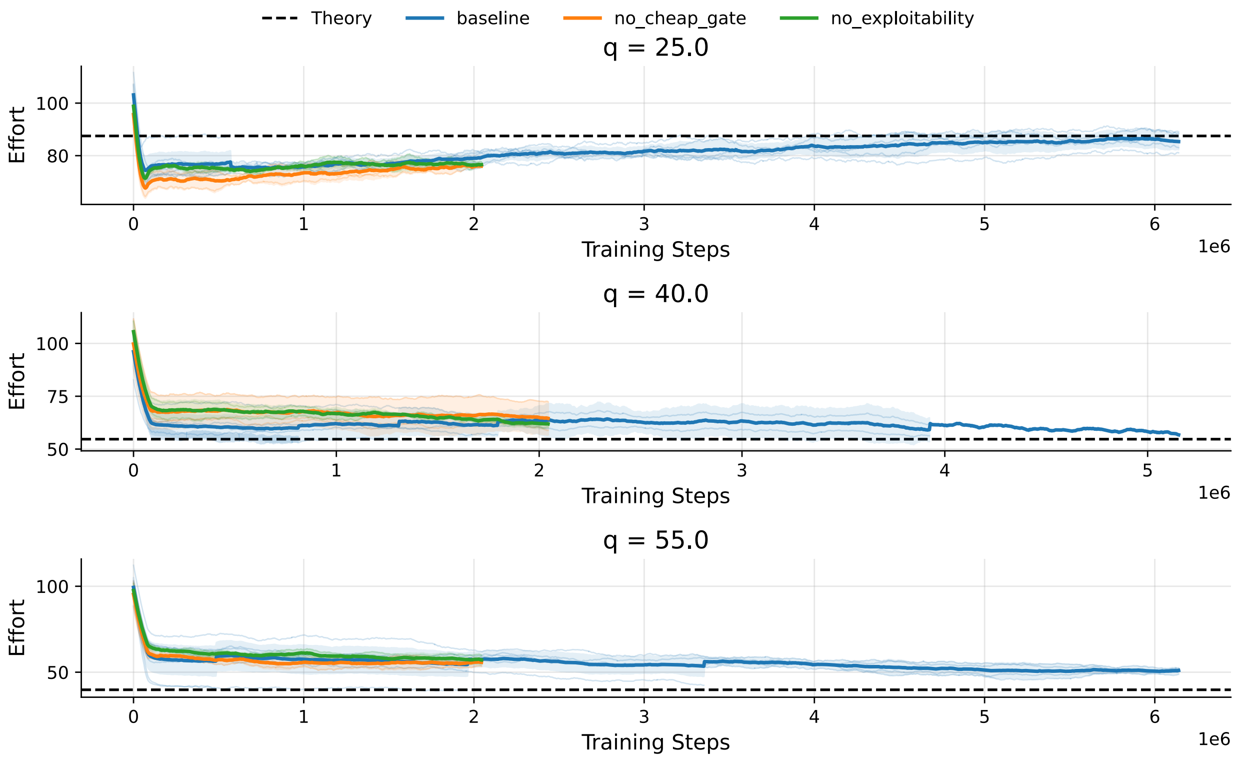
a. Cheap gate 改为 Stability screening passed

b. Nash convergence 改为 Approx. Nash verified

c. Threshold 改为 Tolerance threshold

11- Figure 6b: Exploitability in the excluded low-noise case $q = 25$

12- Component ablation of TEL-PPO across noise levels



a. 统一 y 轴

b. Theory 线更明显些

c. Baseline 改为 TEL-PPO

d. No cheap gate 改为 No stability screening

e. No exploitability 改为 No exploitability verification

f. TEL-PPO 更粗些，ablation 略细一些

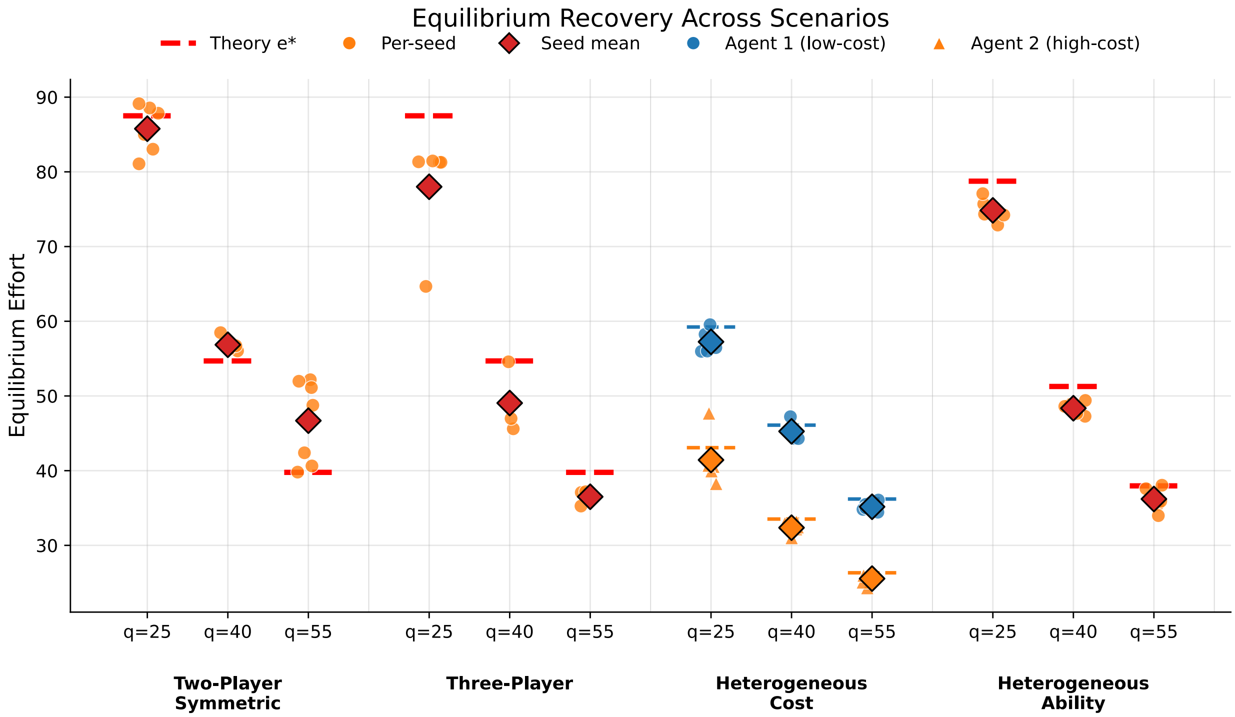
g. Training steps（x10^6）

h. q=35,40,55,去掉.0

i. 是否能加上一个小的 summary：

- terminal absolute error
- final exploitability
- NC rate
- time-to-pass verification

13- Equilibrium Recovery Across Scenarios and Noise Levels



a. Name: Equilibrium Recovery Across Scenarios and Noise Levels

b. 把横轴的分组更清晰：加一个（灰）竖线或者增加空隙？或者 symmetric 和 heterogeneous 两部分用两种背景分区

c. Theory 线增强区分，改为灰黑色？

d. Per-seed 改为 Per-seed estimate; Seed mean 改为 Across-seed mean

e. Per-seed estimate 颜色较淡的小点

f. Scenario 标签更靠近 q 值

14- Table 1. Tournament Environment Configuration

Parameter	Value/Description
Number of players	n = 2 by default; n = 3 in the multi-player variant
Winner and loser prizes	$(w_{H,} w_L) \in (6.5, 3), (8,4)$

Cost parameter	$k \in \{0.0004, 0.0005\}$
Effort bounds	$[e_{min,}\ e_{max}] = [0,200]$
Noise distribution	$q \ \in \ \{25, 40, 55\}$, with $\varepsilon_i \sim U[-q, q]$
Default benchmark scenario	$n = 2;\ w_H = 6.5;\ w_L = 3;\ k = 0.0004$
Variation protocol	One environmental factor is varied at a time around the default benchmark unless a scenario-specific variant is explicitly introduced.
Scenario classes	symmetric two-player benchmark; three-player extension; heterogeneous-cost variant; heterogeneous-ability variant
Analytical benchmark validity	Closed-form benchmark is used only in analytically valid interior cases; cases that violate benchmark validity are treated separately
Additional scenario-specific settings	Exact heterogeneous-cost and heterogeneous-ability parameter values are reported in Section 5.2

15- Table 2. TEL-PPO Training and Verification Configuration

Category	Parameter	Value
Optimization	Optimizer	Adam
Optimization	Learning rate (start →end)	$3\times10^{-4}\rightarrow2\times10^{-4}$
Optimization	Samples per update	?
Optimization	Minibatch size	1024

Optimization	Epochs per update	6
Optimization	Random Seeds N_{seed}	?
PPO / Policy	Clip range (start →end)	0.30→0.15
PPO / Policy	Target KL	0.01
PPO / Policy	Entropy coefficient	0 (omitted)?
PPO / Policy	Discount factor γ	1
PPO / Policy	GAE parameter λ	1?
PPO / Policy	Network architecture	Number of layer and units per layer、activation、shared/separate
PPO / Policy	State input	(w^H, w^L, k, q)
Self-play	Opponent synchronization interval	2 updates
Self-play	EMA coefficient τ	0.20
Self-play	History size	10
Self-play	History sampling probability	0.3
Stability / Verification	KL window	20
Stability / Verification	Mean-KL threshold	0.0045
Stability / Verification	Effort-drift threshold	2.0
Stability / Verification	Exploitability tolerance ε	0.05

Stability / Verification	Evaluation interval (N_eval)	every ? PPO updates
Training budget	Maximum PPO updates / environment steps	?

16- Table 3. Quantitative summary for symmetric two-player and three-player settings

Scenario	q	Method	Mean effort±std	$\lvert\bar{e}-e^*\rvert$	Relative error	Exploitability	Structure Gap	Steps to Convergence
Two-Player	25	Theory	87.50	0.00	0.00%	0.000	0.00	-
Two-Player	25	Gradient	87.24±0.00	0.26	0.30%	-	0.00	-
Two-Player	25	TEL-PPO	85.77±3.27	2.74	3.13%	1.160	0.02	4176555
Two-Player	40	Theory	54.69	0.00	0.00%	0.000	0.00	-
Two-Player	40	Gradient	54.78±0.00	0.10	0.18%	-	0.00	-
Two-Player	40	TEL-PPO	56.86±0.95	2.17	3.96%	0.009	0.01	991232
Two-Player	55	Theory	39.77	0.00	0.00%	0.000	0.00	-
Two-Player	55	Gradient	39.85±0.00	0.07	0.18%	-	0.00	-
Two-Player	55	TEL-PPO	46.69±5.54	6.92	17.40%	0.057	0.01	999424
Three-Player	25	Theory	87.50	0.00	0.00%	0.000	0.00	-
Three-Player	25	Gradient	87.70±0.00	0.20	0.22%	-	0.00	-
Three-Player	25	TEL-PPO	78.00±7.46	9.50	10.85%	1.284	0.00	NC
Three-Player	40	Theory	54.69	0.00	0.00%	0.000	0.00	-
Three-Player	40	Gradient	54.69±0.00	0.00	0.00%	-	0.00	-
Three-Player	40	TEL-PPO	49.05±4.83	5.64	10.31%	0.020	0.00	1183744
Three-Player	55	Theory	39.77	0.00	0.00%	0.000	0.00	-

Three-Player	55	Gradient	39.90±0.00	0.12	0.31%	-	0.00	-
Three-Player	55	TEL-PPO	36.51±1.08	3.26	8.21%	0.024	0.00	NC

17- Table 4 Quantitative summary for heterogeneous tournament settings

Scenario	q	Method	Mean effort±std	$\lvert\bar{e}-e^*\rvert$	Relative error	Exploitability	Structure Gap	Steps to Convergence
Het. Cost	25	Theory	51.15	0.00	0.00%	0.000	0.00	-
Het. Cost	25	TEL-PPO	49.34±2.53	2.80	5.47%	0.118	15.82	NC
Het. Cost	40	Theory	39.81	0.00	0.00%	0.000	0.00	-
Het. Cost	40	Gradient	38.59±0.00	1.22	3.06%	-	3.24	-
Het. Cost	40	TEL-PPO	38.81±0.95	1.11	2.80%	0.026	12.90	NC
Het. Cost	55	Theory	31.27	0.00	0.00%	0.000	0.00	-
Het. Cost	55	TEL-PPO	30.36±0.62	0.91	2.90%	0.026	9.62	NC
Het. Ability	25	Theory	78.75	0.00	0.00%	0.000	0.00	-
Het. Ability	25	TEL-PPO	74.84±1.60	3.91	4.96%	0.839	0.00	NC
Het. Ability	40	Theory	51.27	0.00	0.00%	0.000	0.00	-
Het. Ability	40	Gradient	51.27±0.00	0.00	0.00%	-	0.00	-
Het. Ability	40	TEL-PPO	48.36±0.88	2.91	5.68%	0.026	0.00	NC
Het. Ability	55	Theory	37.96	0.00	0.00%	0.000	0.00	-
Het. Ability	55	TEL-PPO	36.21±1.65	1.79	4.72%	0.026	0.00	NC